

Ethics, Guardrails & Responsible AI

Module 6

Duration: 2 Weeks

Topic 1: Responsible AI Foundations & Risk Awareness

- Understanding bias, hallucination, and misuse risks in GenAI systems.
- Transparency, explainability, and accountability in AI decisions.
- Enterprise expectations for ethical AI adoption.
- AI Act overview — risk tiers, prohibited uses, compliance obligations, data principal rights, consent framework for AI apps

Topic 2: Safety Guardrails & Human-in-the-Loop Controls

- Input and output validation strategies.
- Policy enforcement, prompt restrictions, and fallback handling.
- Human approval loops for high-risk AI actions.
- NeMo Guardrails (colang rules) + Llama Guard 3 (open-source safety classifier)
- Prompt injection detection and defense patterns

Topic 3: Governance, Compliance & Audit Readiness

- Monitoring AI behavior in production.
- Audit logs, traceability, and decision documentation.
- Aligning GenAI systems with enterprise and regulatory standards.
- Model cards and AI system documentation standards

Project 31 LLM Toxicity Monitoring & Safety Guardrail Engine

Responsible AI

Create a safety pipeline to detect toxic content, enforce prompt rules, and generate safer LLM responses.

OpenAI

Perspective API

Rule-based guardrails

Program Electives

Choose any 2 electives

A. GenAI and Agentic AI for **BFSI**

B. GenAI for **Retail & ecommerce**

C. GenAI for **Manufacturing & Automotive**

D. GenAI for **Healthcare & Pharma**

E. GenAI for **Cloud & Devops**

F. GenAI for **Managers & Leaders**

G. GenAI for **Software Developers**

H. GenAI for **QA & Test Engineers**

Gen AI Specialization for BFSI Professionals

Duration: 1 Month

Elective A

BFSI Domain Fundamentals

- KYC → onboarding → transactions → credit → servicing → collections
- Financial services workflows: portfolio advisory → compliance → fraud → risk scoring
- Insurance workflows: claims → underwriting → policy servicing → grievance redressal
- Key BFSI metrics: default rate, fraud probability, risk score, NPA metrics, claim turnaround, compliance accuracy
- RBI AI/ML guidelines for financial institutions — guardrails for AI in lending

Project 1 GenAI Credit Underwriting Document Assistant

Credit Risk

Build LLM assistant that reviews financial documents and generates eligibility summaries for credit decisions.

Azure

LLM Summarization

Document Parsing

Project 2 GenAI Claims Processing Assistant

Insurance

Summarize claim documents, detect missing information, and generate verification checklists for adjusters.

LLM Review

Structuring Templates

PDF Processing

Project 3 Agentic KYC Document Verification Pipeline

Compliance

Multi-agent pipeline extracting, validating, and cross-checking KYC documents automatically.

LangGraph

Vision Models

Rule Engine

Project 4 Agentic Compliance Review Agent

Banking

Build agent that checks policy documents against regulatory guidelines and flags compliance deviations.

RAG

Agentic Workflows

RBI Rule Parsing

Project 5 Agentic Fraud Detection & Explanation Engine

Fraud Detection

Implement multi-agent workflow that detects anomalies, explains fraud likelihood, and drafts investigation reports.

CrewAI

LangGraph

Reasoning Tools

Project 6 GenAI Digital Banking Q&A Assistant

Digital Banking

Create LLM-powered chatbot that answers customer queries using banking FAQs and product details.

LLM Prompting

Retrieval

Conversational Pipelines

Gen AI Specialization for Retail & E-com

Duration: 1 Month

Elective B

Retail & E-com Domain Fundamentals

- Retail workflows:
- catalog → listing → pricing → promotion → order → delivery → returns
- E-commerce stack overview:
- search, recommendation, merchandising, demand planning, customer support
- Key KPIs: conversion, AOV, returns rate, search-to-cart %, fill rate, OTIF

Project 1 Visual Product Search & Description Generator

E-commerce

Build a multimodal app: upload product image → GPT-4o generates title, description, tags automatically.

GPT-4o Vision

LangChain

Streamlit

Project 2 GenAI Returns Insights & Recommendation Assistant

E-commerce

Analyze return reasons using GenAI and generate improvement suggestions for products, logistics, or packaging.

LLM Summaries

RAG

Sentiment Models

Project 3 **Agentic Customer Query Resolution Agent**

Customer Support

Build multi-agent workflow that resolves customer queries, checks order status, and performs automated follow-ups.

Autogen

API Tool Calling

LangGraph

Project 4 **Agentic Catalog Enrichment Workflow**

E-Commerce

Create multi-agent workflow that extracts attributes, enriches product data, and validates outputs automatically.

CrewAI

LangGraph

CSV Tools

Project 5 **Agentic Pricing Recommendation Agent**

Retail

Build an agent that analyzes demand signals and competitor pricing to generate dynamic price suggestions.

LangChain

Data Tool Calls

LLM Reasoning

Project 6 **GenAI Bundle Recommendation Engine**

Growth & Analytics

Generate product bundle ideas using purchase patterns, browsing behavior, and LLM reasoning for higher conversions.

LLM Reasoning

Association Rules

Behavioral Data

Gen AI Specialization for Manufacturing & SupplyChain

Elective C

Duration: 1 Month

Manufacturing & Supplychain Fundamentals

- Manufacturing workflows:
- production planning → machining → assembly → quality → packaging → maintenance
- Supply chain workflows:
- procurement → inventory → warehousing → shipping → logistics → last-mile
- Key metrics:
- OEE, MTBF, defect rate, cycle time, lead time, OTIF, fill rate

Project 1 Assembly Troubleshooting RAG Agent

Manufacturing

Build RAG system that retrieves fault, tool, and assembly instructions to help technicians resolve issues faster.

LangChain

Vector DB

LLM Retrieval

Project 2 GenAI Inventory Insights RAG Assistant

Supply Chain

Build a RAG assistant that answers inventory, stock-level, and replenishment queries using supply chain documents.

LangChain

Vector DB

Azure Search

Project 3 **Agentic QA Inspection Workflow**

Quality Control

Implement multi-agent system where inspectors analyze images, classify defects, and generate corrective actions.

CrewAI

Vision Models

LangGraph

Project 4 **GenAI Logistics RCA Assistant**

Logistics

Use GenAI to analyze delivery failures and generate likely root causes with corrective action suggestions.

LLM Reasoning

Document Embeddings

RCA Prompt

Project 5 **Agentic Warehouse Task Planner**

Warehousing

Build an autonomous agents system that assigns warehouse tasks using SKU priority and worker availability.

CrewAI

LangGraph

Scheduling Logic

Project 6 **GenAI Predictive Maintenance Report Assistant**

Manufacturing

Build GenAI tool that explains sensor anomalies and generates human-readable maintenance recommendations.

LLM Reasoning

Time-Series Extraction

Summarization

Gen AI Specialization for Healthcare Professionals

Duration: 1 Month

Elective D

Healthcare Domain Fundamentals

- Healthcare workflows: patient intake → diagnosis → treatment → documentation → billing
- Pharma workflows: R&D → clinical trials → manufacturing → safety → pharmacovigilance
- Key metrics: ADRs, trial duration, readmission rate, LOS, claim turnaround, safety compliance
- HIPAA/HL7 FHIR basics — privacy and interoperability standards for AI in healthcare

Project 1 Agentic ICD/CPT Coding Assistant

Healthcare

Build agent that extracts diagnosis/procedure codes from clinical notes using rule checks and LLM reasoning.

CrewAI

Rule-Based Validation

RAG optional

Project 2 RAG-Based Drug Safety Signal Investigator

Drug Safety

Build RAG system that identifies adverse drug reactions by retrieving evidence from case reports and literature.

LangChain

Vector DB

LLM Reasoning

Project 3 **GenAI Trial Protocol Summary Assistant**

Clinical Trials

Summarize complex clinical trial protocols into simplified inclusion criteria, procedures, and risk summaries.

JSON Output

LLM Summarization

Document Chunking

Project 4 **Agentic Differential Diagnosis Generator**

Diagnosis Support

Create agent that analyzes symptoms, retrieves guidelines, and suggests possible differential diagnoses safely.

CrewAI

RAG

Safety Filters

Project 5 **GenAI FDA/EMA Compliance Checker**

Compliance & QA

Build system that checks SOP content against regulatory standards and highlights missing compliance elements.

RAG + LLM

Validation Layers

Regulatory Documents

Project 6 **GenAI Patient Information Assistant**

Patient Support

Build chatbot that explains diagnoses, treatments, and care instructions in simple language for patients.

LLM Prompting

Safety Filters

Conversation Flow

Gen AI Specialization for Cloud & DevOps Professionals

DURATION: 1 MONTH

Elective E

Generative AI in DevOps

- Introduction to GenAI in DevOps, AI for IaC automation, AI-assisted CI/CD
- AI-Powered Monitoring & Incident Resolution
- Predictive Analytics & Resource Management
- GenAI for Container Orchestration — Kubernetes manifest generation, Dockerfile
- Disaster Recovery & Security Automation with AI
- Fine-tuning LLMs for DevOps tasks + OpenTelemetry-based LLM observability
- GitHub Copilot / AWS CodeWhisperer — AI coding assistants in DevOps pipelines

Project 1 **CI/CD Pipeline Optimization Assistant**

Cloud

Build GenAI tool that analyzes CI/CD logs, detects bottlenecks, and suggests optimized pipeline steps.

Jenkins

GitHub Actions

LLM Reasoning

Project 2 **Agentic Incident Response & Runbook Execution Bot**

Cloud Ops

Create multi-agent workflow that triages alerts, references runbooks, and recommends resolution actions.

CrewAI

Runbook RAG

Alert Manager Logs

Project 3 AI Log Pattern Summarizer & Anomaly Detector

Observability

Build GenAI pipeline that clusters log patterns, detects anomalies, and generates incident insights automatically.

ELK

Embeddings

LLM Summaries

Project 4 GenAI Security Drift & Compliance Checker

Cloud Security

Detect misconfigurations, compare cloud states with policies, and generate compliance violation summaries.

OPA

LLM Reasoning

Cloud Config APIs

Project 5 AI-Driven Kubernetes Auto-Scaling Agent

Cloud Infra

Agent monitoring cluster metrics and automatically generating + applying HPA/VPA manifests using LLM reasoning.

LangGraph

Kubernetes Python Client

GPT-4o

Prometheus

Project 6 GenAI Cloud Infrastructure Cost Optimizer

FinOps

Analyze cloud resource usage, identify waste, and generate cost reduction recommendations using LLM reasoning.

LLM Reasoning

AWS Cost APIs

LangChain

Gen AI Specialization for Managers & Leaders

Duration: 1 Month

Elective F

Generative AI For Managers

- Agile Foundations & Practices — Scrum, Kanban, Jira, Sprint Planning
- Tools & Practical Execution — Jira, OpenProject, Slack, Teams, Confluence
- GenAI for Digital Business Transformation, AI Governance
- Agent-based Decision Systems, AI ROI Evaluation
- Prompt-based business automation with ChatGPT Enterprise / Microsoft Copilot
- Building an AI business case: ROI estimation, risk-benefit framework for C-suite

Project 1 AI Sprint Planning & Jira Workflow Setup

Agile Delivery

Plan a full AI sprint, create Jira boards, define epics, and break user stories into actionable tasks.

Jira

User Story Creation

Sprint Planning

Project 2 AI Meeting Intelligence & Action Item Tracker

Productivity

Transcribe meetings, extract decisions and action items, assign owners, and sync outputs to project tools.

Whisper STT

LangChain

Jira Integration

Project 3 **GenAI KPI Insights & Forecast Dashboard for Managers**

Business Strategy

Build AI-driven KPI dashboard that summarizes trends, identifies risks, and suggests business actions for leaders.

LLM Summaries

KPI Templates

Forecast Models

Project 4 **GenAI Marketing Campaign Design & Optimization Assistant**

Growth Strategy

Use GenAI to design campaigns, segment audiences, and generate strategic recommendations for managers.

LLM Reasoning

Prompt Templates

Customer Segmentation

Project 5 **AI Risk Assessment & Mitigation Planning Framework**

Compliance

Identify operational and financial risks, generate AI-assisted mitigation plans, and produce risk heatmaps.

LLM Analysis

Risk Scoring Matrix

Compliance Checklists

Project 6 **GenAI Root Cause & Action Plan Generator**

Operations

Use GenAI to analyze incidents, identify root causes, and auto-generate structured action plans for managers.

RCA Tree

LLM RCA Prompts

Workflow Templates

Gen AI Specialization for Software Developers

Duration: 1 Month

Elective G

Topic 1: AI-Powered System Design

- Designing scalable GenAI & agentic applications — context routing, retrieval layers
- Architecture choices for multimodal, RAG, multi-agent workflows
- Latency, cost, safety, reliability considerations

Topic 2: Production Architecture & Infrastructure Patterns

- Event-driven AI patterns, Model gateways, Inference routers, Distributed retrieval
- Monitoring, observability, fallback systems, guardrail enforcement
- FastAPI + Docker + Kubernetes + vLLM serving patterns
- Gateway pattern — single API façade routing to OpenAI / Azure / local vLLM based on cost/latency SLAs

Topic 3: Backend Architecture for LLM Applications & RAG Microservices

- Backend Architecture, High-Performance API (Async, Batching, Streaming)
- Caching, Persistence & Storage for AI Microservices
- Distributed RAG & AI Microservice Patterns
- Semantic caching — cache LLM responses by embedding similarity, not exact key match

Project 1 AI Inference Gateway & Routing Layer

System Design

Create an inference router balancing requests across different LLMs based on cost and performance.

API Gateway

Routing Logic

Project 2 High-Performance LLM API Service

API Architecture

Build a low-latency API service delivering LLM responses using async execution and intelligent routing.

FastAPI

AsyncIO

Project 3 Intelligent Prompt Caching Microservice

Backend Systems

Implement a backend service that caches prompts and responses to reduce repeated computation and system load.

TTL Strategies

Hash Keys

Project 4 Multi-LLM API Gateway with Fallback

System Design

API gateway routing to GPT-4o → Azure fallback → vLLM local, based on latency budget and cost limits.

FastAPI

LiteLLM

Docker

vLLM

Gen AI Specialization for QA & Test Engineers

Duration: 1 Month

Elective H

QA & Testing Domain Fundamentals

- QA workflows: test planning → execution → defect reporting → regression → sign-off
- Testing types: functional, regression, integration, performance, security, UAT — distinct AI use cases
- Test management tools: Jira, TestRail, Zephyr, Xray — AI assistant integration points
- Key QA metrics: defect density, test coverage, pass/fail ratio, escaped defect rate, execution velocity

Project 1 GenAI Test Case Generator from Requirements

Test Design

Build LLM pipeline that reads user stories and generates test cases with priority and coverage.

OpenAI API

LangChain

Jira API

Project 2 AI Bug Triage & Severity Classifier

Defect Management

Classify defect severity, extract root cause signals, and auto-route bugs using LLM pipelines.

LLM Classification

RAG

Vector DB

Project 3 **Agentic Regression Test Suite Selector**

Test Optimization

Build an agent that analyses code diffs, maps affected modules, and selects the regression suite.

LangGraph

GitHub API

Embeddings

Project 4 **GenAI Test Automation Script Generator**

Automation Engineering

Generate Playwright or Pytest scripts from plain-language test descriptions with self-healing locator suggestions.

OpenAI Code Gen

Playwright

Agentic Loop

Project 5 **Agentic QA Coverage Gap Analyst**

QA Strategy

Create a multi-agent workflow that maps test suites to requirements and generates a coverage report.

CrewAI

RAG + Vector DB

LLM Reasoning

Project 6 **GenAI Sprint Quality Insights Reporter**

Reporting & Analytics

Ingest test execution logs and defect data to generate sprint quality summaries and trend analysis.

LLM Summaries

LangChain

TestRail / Jira API

Capstone Projects

1

GenAI Quality & SOP Intelligence Platform

Summarize SOPs, analyze production logs, and generate real-time quality insights for defect prevention.

LLMs

LangChain

RAG

Azure OpenAI

2

GenAI Vehicle Diagnostics & Troubleshooting Assistant

Retrieve service bulletins and generate fault explanations with repair suggestions using domain RAG.

RAG

LLM Reasoning

Diagnostic Manuals

3

Agentic Logistics Optimization Orchestrator

Agents optimize delivery routes, analyze delays, and automate logistics scheduling.

Autogen

CrewAI

Routing APIs

4

Agentic Catalog Management & Dynamic Pricing Agent

Autonomous agents enrich catalog data, fix attributes, and recommend pricing strategies.

LangGraph

Autogen

Pricing Rules

Capstone Projects

5

Agentic Fraud Detection & Investigation System

Agents analyze anomalies, detect fraud patterns, and generate investigation summaries.

Autogen

Workflow Agents

Rule Engines

6

Agentic Clinical Workflow Orchestrator

Agents support triage, coding, documentation, and clinical workflow automation.

RAG

LangGraph

Autogen

7

GenAI Clinical Trial & Drug Safety Insights Platform

Summarize trial protocols, extract ADRs, and generate domain-specific safety insights.

RAG

LLMs

Scientific Parsing

8

GenAI Cloud Log Analyzer & RCA Generator

Analyze logs, detect anomalies, and produce RCA explanations via log-aware LLM pipelines.

Log Parsing

LLMs

Runbook RAG